



Dr. Mujtaba Ellahi

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Career objective

- A researcher that needs experiences in polymers and composite materials possess attractive range of properties for the development of superior engineering applications. Such as polymer synthesis and characterization, composite materials design and manufacture and characterization, Poly thiol and poly amine based polymer disperse liquid crystal films with electro-optical properties and morphology, liquid crystal materials and blue phase liquid crystal materials, reflective color E-paper, light selective reflected and transmitted films, radiation cross linking and degradation polymers.
- Nano-silver powder has become a new engineering material with possible applications in many fields; it is interesting to study the role of nano particles into the PDLCs morphology and their electro-optical properties. The influences of doping different concentrations of nano-Ag on the diffraction efficiency, driving voltage and response time will investigate by measuring optical and electro-optical properties of the PDLCs films.
- A Chemistry teacher position that demands proficiency in advanced Chemistry (polymer, analytical, organic and industrial) with solid theoretical background and excellent experimental skills, high passion toward teaching

Education

- | | | |
|-------------------------------------|--|---|
| ➤ Ph. D. | Materials Science and Engineering | Effects of polymeric network on PDLC films |
| ✧ | University of Science and Technology Beijing | 85.3% A-1. |
| ➤ M. Sc.(Honours) | Chemistry | Organic Chemistry |
| ✧ | University of Sindh, Jamshoro | ~ 1 ST Division |
| ➤ B. Sc.(Honours) | | 2 ND Division Chemistry |
| ➤ H.S.S.C. (Pre-Engineering) | | A-Grade. Chemistry, Physics, Math. |
| ➤ S.S.C. | | A-Grade. Chemistry, Physics, Math. |

Research Experiences

- **Preparation of PDLC films with epoxy resin**
 - ✧ The influence of the excessive mercaptane hardener effects of epoxy monomer structure on the electro-optical properties and morphology of polymer-dispersed liquid crystal films.
 - ✧ Detailed analysis and finding relation between droplet size/shape of liquid crystal and properties using the hardener (Polyamine and Poly thiols) of epoxy based PDLC films.
 - ✧ Influence of enhanced curing temperature of epoxy monomers structure on the electro-optical properties and morphology of polymer-dispersed liquid crystal films.

- ✧ The study of di, tri, tetra and multifunctional epoxy monomers structures.
- ✧ Investigation of step growth addition polymerization mechanism reaction of PDLC films.
- ✧ Coin like and small hole domain size of liquid crystal studied.
- ✧ Investigation of the controlled droplet size of liquid crystal with heat curing PIPS method.
- ✧ Explaining and understanding the effects of cross linked polymer network of PDLC films.

Analysis techniques:

- ✧ Liquid crystal device (LCD) parameters tester
- ✧ Scanning Electron Microscopy (SEM)
- ✧ (VIS) region using UV-VIS-NIR spectrophotometer
- ✧ Abbe Refractometer
- ✧ Infrared spectrometer
- ✧ Polarized Optical Microscopy (POM)

PUBLICATIONS

[1] **Mujtaba Ellahi**, et al, Influence of the multi-functional epoxy monomers structure on the electro-optical properties and morphology of polymer-dispersed liquid crystal films [J].Polymer Bulletin, 2013, 70 (11): 2967-2980. (ISI Impact Factor: 1.332).

[2] **Mujtaba Ellahi**, et al.,Influence of the composition of the curable epoxy monomers structure on the characterization and morphology of polymer-dispersed liquid crystal films [J].Soft Materials, (**Accepted**) (ISI Impact Factor: 1.429).

[3] **Mujtaba Ellahi**, Yanzi Gao, M. Y. Rafique, Dil Faraz Khan and Cao Hui. Influence of the excessive mercaptane hardener effects of epoxy monomer structure on the electro-optical properties and morphology of polymer-dispersed liquid crystal films [J].Archives Des Sciences, 2013, 66(4):451-468 .(ISI Impact Factor: 0.536).

[4] **Mujtaba Ellahi**, Fang Liu, Hui Cao and Huai Yang. Effects of the content of Epoxy Monomers on the Electro-Optical Properties and reducing morphology of Polymer-Dispersed Liquid Crystal films. International Conference on Engineering and Applied Sciences (ICEAS-2012) Beijing.759-767.ISBN 978-986-87417-1-3.

[5] Ping Song, Hui Cao, Feifei Wang, Fang Liu, Jingjing Wang,**Mujtaba Ellahi**,Fasheng Li and Huai Yang. The UV polymerization temperature dependence of polymer dispersed liquid crystals based on epoxies/acrylates hybrid polymer matrix components [J].LiquidCrystals, 2012, 39(9): 1131-1140 .(ISI Impact Factor: 1.959).

[6] **Mujtaba Ellahi**, et al, Influence of the multiple crack growth effect of Epoxy Monomers Structure on the Electro-Optical Properties and morphology of Polymer-Dispersed Liquid Crystal films.The 2ND International Conference on Engineering and Applied Sciences (ICEAS-2013) Tokyo.806-815.ISBN 969-986-87416-2-4.

[7] Ping Song, Hui Cao, Feifei Wang, **Mujtaba Ellahi** and Huai Yang.Study of polymer dispersed liquid crystal systems using epoxies / acrylates as hybrid polymer matrix components [J]. Liquid Crystals, 2012(8):39,903-909. (ISI Impact Factor: 1.959).

[8] **Mujtaba Ellahi**, et al, Effects of di and tetra functional epoxy monomers structure on the morphology and the electro-optical properties of polymer-dispersed liquid crystal films [J] International Journal of Scientific & Technology Research, 2013, 2(3):127-131. (Impact Factor: 0.675).

[9] M.YasirRafique, LiqingPan, WaheedSKhan, M.Zubair Iqbal, Hongmei Qiu, M. Hassan Farooq, **Mujtaba Ellahi** and zhengangGuo.Controlled synthesis, phase formation, growth mechanism, and magnetic properties of 3-D CoNi alloy microstructures composed of nanorods [J].Crystal Engineering Communication, 2013, 15(48):5314-5325.(ISI Impact Factor: 3.879).

[10] **Mujtaba Ellahi**, et al, Study of the effects of epoxy monomer structure on the morphology of crosslinked polymer networks and the electro-optical properties of polymer-dispersed liquid crystal films [J].Journal of Polymer Research, (**Under Review**) (ISI Impact Factor: 2.019).

[11] M Zubair Iqbal, Fengping Wang, Aiguo Wu, M.Yasir Rafique, Shujjat Ali, M. Hassan Farooq and **Mujtaba Ellahi**.Hydrothermalsynthesis, characterization and hydrogen storage of SnS nanorods [J].Material Letters, 2013, 106(01):33-36. (ISI Impact Factor: 2.224).

[12] **Mujtaba Ellahi**, Yanzi Gao and M. Y. Rafique. Influence of enhanced curing temperature of epoxy monomers structure on the electro-optical properties and morphology of polymer-dispersed liquid crystal films [J].American Journal of Engineering Research, 2013, 2(3):01-06.

[13] M.YasirRafique,Liqing Pan, M. Zubair Iqbal, Rafi-ud-din, Hongmei Qiu, M. Hassan Farooq, Zhengang Guo, **Mujtaba Ellahi**. Fabrication of CoNi alloy hollow-nanostructured microspheres for hydrogen storage application [J]. Journal of Nanoparticle Research, 2013, 15(1768):1-11. (ISI Impact Factor: 2.175).

[14] LIU Fang, CAO Hui, JI Peng-fei, LIU Kai-qiang, **ELLAHI Mujtaba**, YANG Zhou and YANG Huai. Preparation and Electro-Optical Properties of Polyurethane Matrix Based Polymer Dispersed Liquid Crystal Films [J].Chinese journal of liquid crystals and displays, 2013, 28(1):1-6.

[15] Murad Ali Khaskheli, Ping Wu, Amir Mahmood Soomro, Matiullah Khan, **Mujtaba Ellahi**, Asadullah , Mohammad Siddique Kalhoro. Fabrication and Electrical characteristic of quaternary ultrathin HfTiErO thin films for MOS devices grown by rf sputtering [J]. Chemistry and Materials Research, 2013, 03(04):41-47.

[16] Ling Wang,Dongyu Zhao,Wanli He,Qian Wang,Huai Yang,Meina Yu,and , **Mujtaba Ellahi**,Lin Guo Xia Xiao,Yang Zhang. Polymer-stabilized nanoparticle-enriched blue phase liquid crystals[J].Journal of Material Chemistry C, 2013, 40 (01): 6526-6531. (ISI Impact Factor: 6.101).

[17] Murad Ali Khaskheli, Ping Wu, M S Kalhoro, H. NizamaniI, **Mujtaba Ellahi**, Matiullah Khan, Asadullah ,. M. Z. Iqbal, D.F. Khan, M. Yasir. Electrical and Photonics Characteristics of Hftiero Thin Films Deposited at 673 K Substrate Temperature [J] Sindh University Research Journal Sci: Series, 2013, 45(02):317-320.

[18] Dil Faraz Khan, Haiqing Yin ,He Li,Zainul Abideen ,Asadullah , Xuanhui Qu ,**Mujtaba Ellahi** . Effect of impact force on TiC10Mo alloy powder compaction by high velocity compaction technique [J] Materials and Design, 2013, 54(02):149-153. (ISI Impact Factor: 2.913).

CONFERENCE PROCEEDINGS

1. **Mujtaba Ellahi**, Fang Liu, Hui Cao and Huai Yang. Effects of the content of Epoxy Monomers on the Electro-Optical Properties and reducing morphology of Polymer-Dispersed Liquid Crystal films. **International Conference on Engineering and Applied Sciences (ICEAS-2012) Beijing**.759-767.ISBN 978-986-87417-1-3.
2. **Mujtaba Ellahi**, et al, Influence of the multiple crack growth effect of Epoxy Monomers Structure on the Electro-Optical Properties and morphology of Polymer-Dispersed Liquid Crystal films. **The 2ND International Conference on Engineering and Applied Sciences (ICEAS-2013) Tokyo**.806-815.ISBN 969-986-87416-2-4.

Research Profile

Impact Factor	27.201
Number of Citations	13
h Index	02

Teaching Experience

- ✧ Served as a **Chemistry Lecturer** in Public College Mirpurkhas Pakistan, 2000- 2010.
- ✧ **Assistant Professor** (IPFP) at Department of Chemical Engineering, University of Karachi 4th Sept: 2013 to Current.

Scholarships and Awards

1. 3rd position in excellent international student scholarship of USTB for year of 2012.
2. Chancellor Scholarships of University of Science and Technology Beijing for Ph.D Studies.
3. Rotary club International scholarship of youth merit award programme.
4. Tharparkar boy scouts presidential scholarship.

Extra abilities

1. Hafiz –ul-Quran.
2. Good orator.
3. Playing Cricket.

Computer skills

- MS. OFFICE (word, excel, PowerPoint)
- Adobe Photoshop(7.0)
- Adobe illustrator
- MDI Jade
- OriginLab (8.0)
- Digital micrograph
- Chem:Draw Ultra

References

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